Sergio Ferniza Marquez

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SUMMARY

Data-oriented professional with hands-on experience building end-to-end data systems, predictive models, and interactive dashboards. Proficient in SQL, Python, C#, Java, Tableau and Power BI, with a strong foundation in database design and cloud environments like Azure and AWS. With proven abilities to derive actionable insights from data and optimize data pipelines in solo and team settings.

EDUCATION

University of California – San Diego
B.S., Mathematics – Computer Science March 2025

- Studied abroad for a year at Lund University, Sweden.
- Relevant coursework: Data analysis and inference, Database systems and optimization, Applied machine learning, Statistical modeling, Forecasting and Stochastic processes.

RELEVANT EXPERIENCE

CODEG, Tijuana, Mexico
Data Specialist (Remote) March 2025 – Present

- Developed and implemented a comprehensive remote data infrastructure to track employee records, transactions, and event costs, ensuring full transparency for this publicly funded nonprofit.
- Built predictive models and conducted forecasting analyses to design data-driven marketing strategies, reducing outreach costs while expanding event reach and engagement.
- Created financial and operational visualizations to support cost analysis, marketing decisions, and resource planning for the organization's youth development events.
- Led cost analysis initiatives to streamline event expenditures, helping optimize budget allocations without compromising program quality.
- Contributed on-site to the organization and execution of professional development events, integrating operational insights with real-time initiative.
- Worked with tools such as Excel, PowerPoint, SQL, Power BI, Python and PostgreSQL

PROJECT EXPERIENCE

Logistics Database Project
Tools: Python, PostgreSQL, Jupyter Notebook, Tableau
Simulated and analyzed end-to-end logistics operations over a multi-year period, including database design, behavioral automation, data generation, and advanced analytics.

- Designed and implemented a relational database to simulate logistics workflows, including inventory management, deliveries, financial logs, employee data and payroll.
- Built python modules to automate operational behaviors (FEFO, first expired first out, order fulfillment, restocking, refueling, and payroll execution) across a multiyear simulated timeline.
- Generated synthetic data for inventory, transactions, and deliveries, and then used ARIMA forecasting to optimize restocking schedules, minimizing stockouts and waste, increasing net revenue.
- Developed a tableau dashboard to visualize monthly costs, revenue, and performance KPIs, allowing for insights into cost drivers and workforce efficiency.
- **GitHub Link** | **Tableau Dashboard Link**

Hospital Simulation & Analytics Project
Tools: Python, PostgreSQL, Jupyter Notebook, pandas, matplotlib, scikit-learn
Simulated and analyzed hospital operations over a multi-week period, including database design, patient behavior modeling, billing automation, inventory management, fraud detection and predictive analytics.

- Designed a fully normalized relational database to model hospital workflows, including patient admissions, treatments, prescriptions, payroll, inventory restocking, and fraud detection.
- Built python modules to automate operational events (hourly admissions, treatments, supply usage, restocking, payroll execution) for over 70,000 patient interactions and tens of thousands of transactions.
- Generated synthetic medical data and injected controlled anomalies in prescriptions and supply chains, identifying well over $32,400 in fraud losses through anomaly detection techniques, while also identifying the culprits of said fraud.
- Applied linear regression models to forecast and visualize monthly revenue and inventory needs, achieving an 18% cost savings with dynamic restocking strategies in comparison to fixed restock schedules.
- Conducted exploratory data analysis and visualizations on patient outcomes by hospital, diagnosis, race, and age, observing equity gaps and areas for quality improvement.
- **GitHub Link**

AirBnDB
Tools: C#, .NET Framework, Windows Forms, Azure, Microsoft SQL Server
With a Swedish team during a study abroad program we developed a complete desktop platform for managing AirBnB-style bookings, from end-to-end. Developing the necessary data infrastructure, intuitive and accessible methods to input new entries and necessary modules to ensure testing, effective architecture and functionality of operations.

- Developed backend logic using C# and SQL procedures, integrated with front-end WinForms components.
- Implemented fill CRUD operations for hosts, users, properties, and bookings through custom data adapters and UI-driven forms.
- Designed and tested modular data access architecture, with secure query execution and automated ID generation.
- Populated our SQL database, with generated listings, proprietors, bookings and transaction logs, to simulate the expected behavior of our platform.
- GitHub Link

Housing Value Trend Analysis
Tools: Python, SQL, Jupyter Notebook, Pandas, Matplotlib
With a team we conducted thorough end-to-end analytics on real world data for housing records, to identify particular points of impact in the housing markets or major Californian metro areas.

- Scraped, cleaned, and conducted EDA of over 13 million housing records of the last 12 years, to study and understand rising prices of the housing markets in neighborhoods across San Diego, Los Angeles and San Francisco.
- Conducted a time series trend analysis on home prices and visualized price distribution, and outliers in affected areas.
- Correlated housing prices with income trends, geographic variables. Such that we could identify patterns that could predict future housing price increases in certain neighborhoods.
- Developed an optimized local dataset by parsing and restructuring identifiers and aggregating long-term value trends.
- GitHub Link |

SKILLS

- Programming languages & tools: Python, Java, C#, SQL, PostgreSQL, Jupyter, MatLab, Azure, AWS & Git.
- Data analysis: Data scraping, Data cleaning, Exploratory data analysis (EDA), Predictive modeling, Time series forecasting (ARIMA), classification (Random Forest), Regression (LASSO), Excel & Cost analysis.
- Database design: Relational schema modeling, Stored procedures, CRUD operations, Data normalization & ETL pipelines.
- Visualizations & Reporting: Tableau, PowerBI, Seaborn, Matplotlib, KPI Dashboards, PowerPoint.
- Language: English (Fluent), Spanish (Native).